OTHER WAYS TO REPRESENT A KNOWLEDGE GRAPH

Resource Description Framework (RDF) is a popular format to represent a knowledge graph. RDF uses subject-predicate-object triples to create a web of linked data, where relationships are expressed as URIs rather than labeled edges as shown below for the graph in figure 5.7 in Turtle format. You can visualize this graph by heading to https://www.ldf.fi/service/rdf-grapher and pasting the RDF below into the box provided and clicking 'Visualize'.

```
@prefix hotel: <http://dtia.com/hotel/>.
@prefix rec: <http://www.realestatecore.io/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

hotel:Hotel rdf:type rec:Building.

hotel:Level1 rdf:type rec:Level ;
             rec:isPartOf hotel:Hotel.

hotel:Level2 rdf:type rec:Level ;
             rec:isPartOf hotel:Hotel.

hotel:Room201 rdf:type rec:Room ;
              rec:isPartOf hotel:Level2.

hotel:Thermostat24967 rdf:type rec:Equipment ;
                      rec:locatedIn hotel:Room201 ;
                      rec:wifiSignalStrength "-70dBm" ;
                      rec:batteryPercentage "83%".
```

The *Web Ontology Language (OWL)* is built on top of RDF and provides a richer vocabulary and constructs to define ontologies such as the *Semantic Sensor Network (SSN)* ontology (https://www.w3.org/TR/vocab-ssn/) for describing sensors and their observations. The DHT22 temperature and humidity sensor that is used in appendix B is defined in the SSN ontology here https://www.w3.org/TR/vocab-ssn/#dht22-description.

Next Generation Service Interfaces NGSI-LD is a graph-based context information model and API based on RDF, OWL, and JSON-LD that provides a uniform representation of entities and attributes as graph-based